SERVER



OCP NIC 2026: Gen 7 and Liquid Cooling – More Speed and Power Ahead

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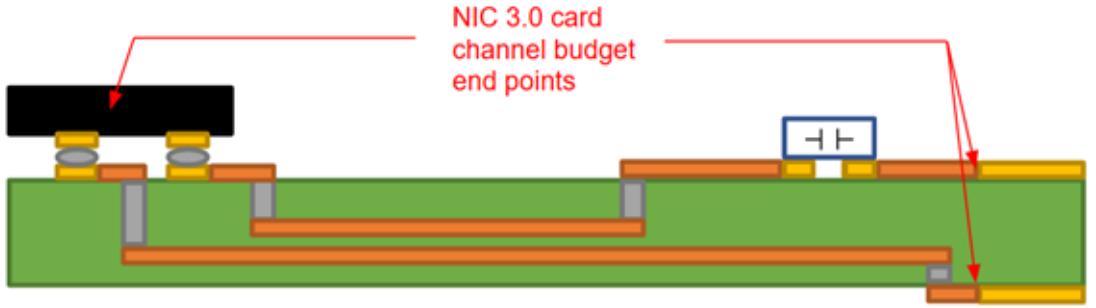


Agenda

- PCIe® Gen 7.0 Support
- PCIe® Fixtures
- Systems Management / FRU Model Updates
- OCP NIC Liquid Cooling Reference Design
- Short Term Power Transients
- 200 W Connections

PCIe® Gen 7.0 Support

- OCP NIC v1.7 introduces PCIe Gen 7.0 support
- Adding Gen 7 enables 1.6 Tbps throughput
- OCP NIC's signal integrity channel insertion loss limits and informative limits will be aligned with the upcoming PCI SIG CEM 7.0 specification
- An improvement vs prior gens where the OCP NIC spec required tighter channel loss saves OCP NICs from using costly extreme ultra low loss PCB materials



Benefit

Supports all common OCP NIC use cases:

- DC-MHS HPM/motherboards
- OEMs (non-DC-MHS) systems
- Cabled solutions

Note: The PCIe® design mark is a registered trademark and/or service mark of PCI-SIG.

PCIe® Gen 7 Fixtures



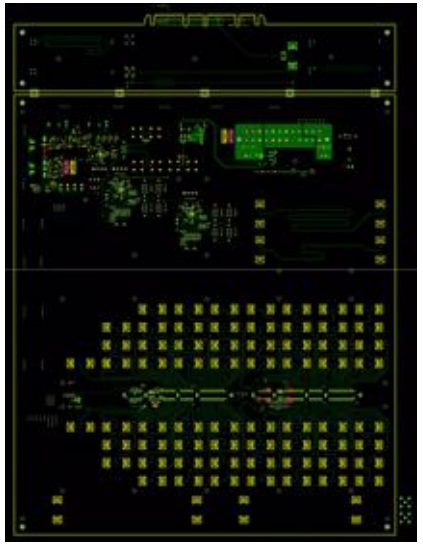
Compliance Load Board

- OCP NIC defines test fixtures used for compliance testing
  - Compliance Load Board (CLB) – test systems with OCP NIC 3.0 slots
  - Compliance Base Board (CBB) – test OCP NIC 3.0 mezzanine cards
- Emulate PCIe CEM CLB & CBB fixtures
- Leverage PCIe test equipment automation without significant modification

- Standardized OCP NIC thermal test fixture
- Firmware available on the OCP Github
  - https://github.com/opencomputeproject/ocp-svr-nic3-ttf-mcu



DSFF Thermal Test Fixture



Compliance Base Board

Systems Management / FRU Model Update

- Enable OCP NIC adoption of the DSP0220 DMTF FRU Format
  - Maintain backwards compatibility with the IPMI FRU model
  - IPMI specification is no longer maintained.
  - DSP0220 already in use by the DC-MHS sub-project.
  - Work with DMTF Manageability representatives for validation tools
- Changes:
  - Board Info Area Mfg Date/Time field - ambiguous mfg date after Nov 24th, 2027.
  - Three-byte field rolls over after this date ⇒ set to 00_00_00h
  - Add DMTF OEM Record (0xC0, IANA 0xB41A00) w/offset to DMTF FRU records

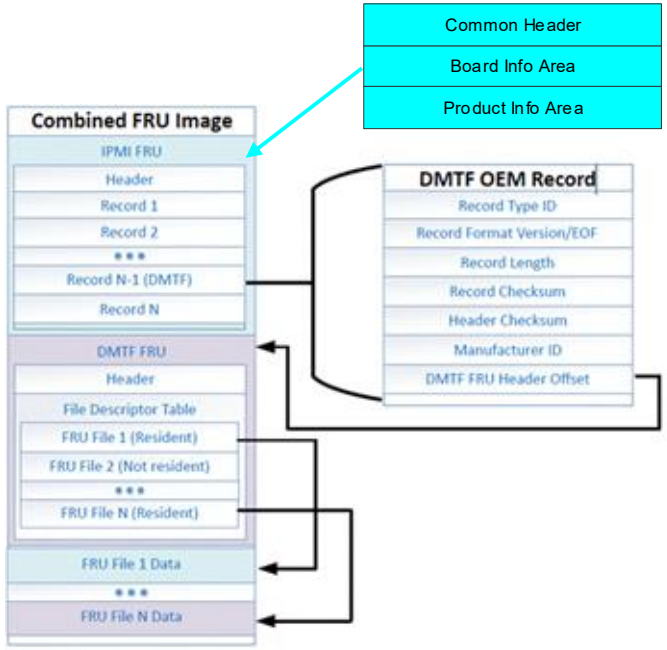
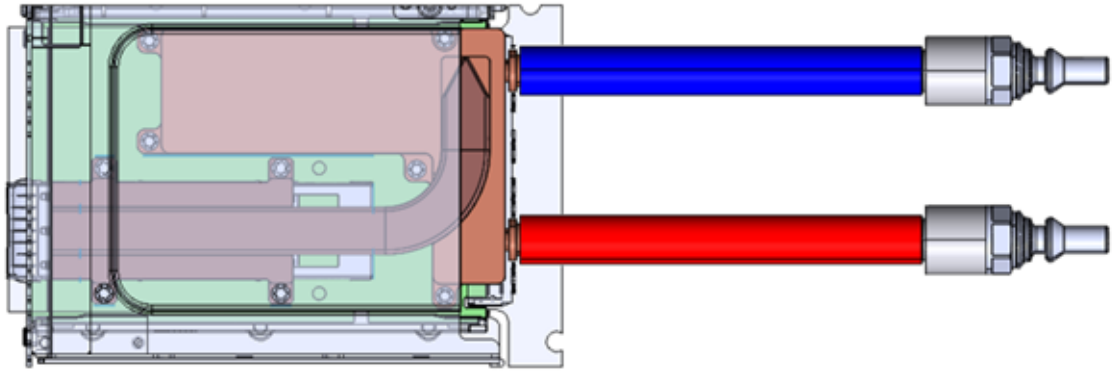


Figure 2 — Combining IPMI and DMTF FRU records

Reference DMTF DSP0220: https://www.dmtf.org/dsp/DSP0220

OCP NIC Liquid Cooling Reference Design

- System liquid cooling will become more common in data centers
- The OCP NIC thermal workstream is developing liquid cooling design options
- Requires solid faceplates
  - No air perforations
  - Ensure ~100% heat captured in liquid
- When contributed, these reference designs will be tied to the OCP NIC specification





Short Term Power Transients

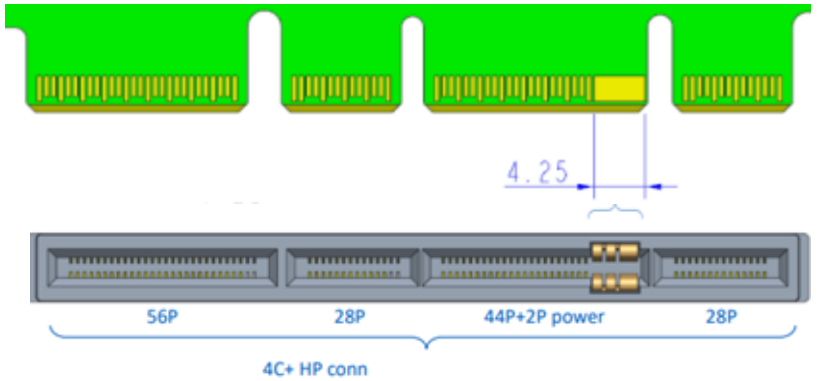
- Enable framework for supporting short term power transients
  - Brings parity with CEM specification
  - Supported in the SFF-TA-1002 1.6a specification with existing connectors

- Updates to the OCP NIC 3.0 specification
  - Change 0xC0 Base OEM Record offset 30 to “Power Excursion Indication”
  - Platform BMCs to parse and understand new NIC power requirements

Reference SNIA public document for transient support: https://members.snia.org/document/dl/55220

200 W Connections - Active Development

- Why:
  - NICs/add-in-cards power demands are increasing
  - 80 W was enough for our traditional NICs
  - Power delivery is limited by the TA-1002 connector
- Need:
  - Higher power connector
  - Backwards compatibility with existing OCP NIC cards.
  - “SFF-TA-1037 4C+ EP” connector as used on the PMM
    - HB thicknesses 1.57 mm to 3.05 mm ← need HB thickness aligned w/ SFF-TA-1002.
    - AIC thickness 1.57 mm
- What needs to change:
  - Prevent TA-1037 card from powering up in TA-1002 connector.
    - Hardware interlock pin (allocate use RFU pin)
    - BMC queries FRU for max power in aux and main power modes
  - Coupled with liquid cooling implementations



Call to Action

- Problems to solve:
  - Enable a faster, smarter and more thermally capable NIC design
    - Enable Gen7 / 1.6T support with OCP NIC
    - Higher power requirements
    - Enable framework for liquid cooling
  - What does a new OCP NIC form factor look like?
- Visit us in the Innovation Village in the “IT Infra” area!
  - Meet with the experts and discuss your OCP NIC needs
- Get involved! Join our monthly calls, and workstream meetings
  - Project Wiki with latest specification: http://www.opencompute.org/wiki/Server/NIC
  - Mailing list: https://ocp-all.groups.io/g/OCP-NIC

Link to OCP NIC project Wiki

Thank You!

